## Quotation No : Q23000393

Date : 31-Aug-2023

Sun Hospital
Tulsipur
CUTTACK
ORISSA

## Kind Attention : Mr./Ms. Dr. Dipesh

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

## For Absstem Industrial Solutions Pvt. Ltd.

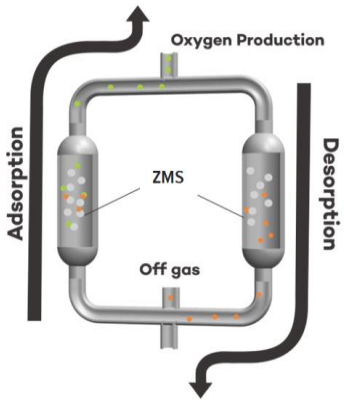
Mohan Reddy
Regional Sales Manager
9289722207
www.absstem.com

medo
POWERED BY ABSSTEM HEALTHCARE

# Medical Grade Oxygen Generator




## PSA Technology




PSA Process explaining oxygen production

Medical Oxygen Generator mainly consists of two adsorbers filled with zeolite molecular sieves. Pressurized air is passed through one of these adsorbers, zeolite having a property to attract nitrogen, adsorbs Nitrogen from air and oxygen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Oxygen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of oxygen produced is monitored by an Oxygen analyser.

## Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid oxygen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid oxygen
- Longevity: at least 10 years

***Absstem Technologies***
***Corporate Address:*** *Plot No. F-591, Sector 8, Phase IV*
*IMT Manesar, Gurugram, Haryana-122051*
*Contact-1800 3010 3394, Email- medo@absstem.com*
*CIN -AAI-2877*

## Description

| | |
|---|---|
| a. Product Name | Oxygen Gas Generator |
| b. Model Range | **MEDO** |
| Technology | PSA (Pressure Swing Adsorption) |

## Specifications

| | | |
|---|---|---|
| **a.** | Oxygen Flow Rate | As per Performance Sheet |
| **b.** | Oxygen Purity | As per Performance Sheet |
| **c.** | Oxygen Delivery Pressure | 5.5 Barg (Higher Pressure Options Available) |
| **d.** | Ambient conditions | |
| | Temperature | 0 to 50 Deg. Celsius |
| | Relative Humidity | up to 80 % (Max.) |
| | Altitude | up to 1000 m above Sea level |

## Scheme Offered- (PSA Package)

This scheme will comprise of –

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Medical Oxygen generator
- Storage Tank
- Oxygen Analyzer
- Alarm panel

## Unique Features of absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for oxygen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Oxygen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

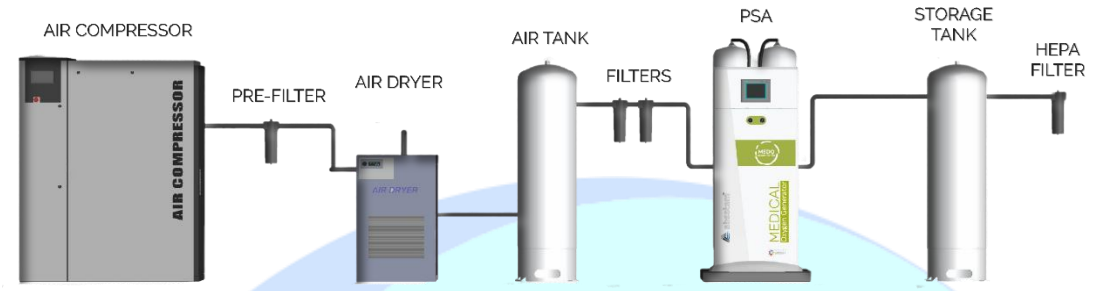
## **Onsite Gas Generation** vs Liquid Tank/Cylinder Alternative

| |
|---|
| *Non-stop availability: 24 hours a day, 7 days a week* |
| *Easy integration within existing installations* |
| *Your own independent supply of medical gas/ reduce your oxygen cost up to **70 to 80%*** |
| *Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses* |
| *No safety hazards when handling high-pressure cylinders* |
| *Easy integration within existing installations* |

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## Equipment Description

The Medical Oxygen Generator consists of




| | |
|---|---|
| Air Compressor | An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **oxygen** from the air. |
| Refrigeration dryer | Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air. |
| Air Tank | Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system. |
| Multistage Filtration system | Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels. |
| PSA system | Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Oxygen** from Air. |
| Storage Tank | Medical Grade **Oxygen** produced from MEDO is stored in this tank to supply a continuous flow of **Oxygen**. |
| Oxygen Analyzer | Oxygen produced is monitored by an Oxygen analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode. |

## HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

### Absstem medical oxygen generator complies with all medical standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation




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## Additional upgrades

- Digital flow meter with totaliser
- High-pressure oxygen tank
- Cylinder manifold and alarm management system
- Remote monitoring

## Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

## Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

## International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and medical directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of medical oxygen generators.

## Itemised Supply description

| No. | Item description | Make | Qty |
|---|---|---|---|
| a | Air Compressor with Filters | KAESER (Germany/IR USA) | 1 no. |
| b | Refrigeration Dryer | KAESER (Germany/IR USA) | 1 no. |
| c | Air Tank | ABSSTEM | 1 no. |
| d | Set of Filters | ABSSTEM | 1 no. |
| e | PSA unit (MEDO) | ABSSTEM | 1 no. |
| f | Electrical Power Panel with PLC and cabling | ABSSTEM & MITSUBISHI | 1 no. |
| f | Digital Oxygen Analyzer | ABSSTEM | 1 no. |
| g | Storage Tank | ABSSTEM | 1 no. |
| h | Outlet product gas Filter | ABSSTEM | 1 no. |

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## Project Site-Completion

We provide Drawing and Layout with complete specifications of Civil (Plant Room), Electrical (Power Supply) and CODPS (Pipeline) requirements, post Site-Inspection. It's Hospital responsibility to complete the installation site before dispatch at his own expenses & to confirm the Site-Completion on time, whereas our engineering team will be available round the clock for assistance at every step of the Site-Completion.

1. Selection of the Installation-Site: The Hospital has to finalize the installation site ether indoors or outdoors before the site engineer inspection as per the space requirement. We can design the Plant as per space availability as well.

2. Base-Flooring for Plant: The Hospital has to complete the flooring job as per the dimensions mentioned in drawing and layout of at least 5 to 10 inches high from the ground level to avoid impact of water logging and minimize vibration effect.

3. Plant Room: The Hospital has to construct a ventilated Plant room as per the Drawing and Layout, to protect the unit from Rain/Birds and to maintain the room temperature during peak summer. An existing similar setup would work as well.

4. Electrical Requirements: The hospital has to complete all the electrical requirements like Electrical Cables, MCCB's, UPS, 3-Phase supply with consistent voltage (415V)/SERVO in case of voltage fluctuations, Dual Exhaust System (18"), Earthing/Grounding, and Power Back-Up (DG) Etc.

5. Training of Personal: The Hospital has to submit a list of personnel for operational training at the time of installation and commissioning.

6. Unloading, Lifting & Positioning: The Hospital has to arrange a Crane for unloading and lifting as well as heavy labors for positioning of the plant with own expenses.

7. Transportation/Freight Charges: The Hospital has to bear the Crane/Carrier charges to the service provider on completion on the job.

## INSTALLATION & COMMISSIONING:

The hospital has to confirm the readiness of site to dispatch the unit. The "Medical Oxygen Plant" would be supplied on a skid mounted system, ready for the installation. Our Installation and Commissioning engineer will visit before arrival of the Plant for the site inspection, assistance in plant unloading, positioning, and installation-commissioning, performance check- up and training of the operators for operation and maintenance. We will provide you detailed operational manual for the unit post Installation & Commissioning.

## Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from Rittal are IP-55 rated makes electrical enclosure internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC.

## Comprehensive Maintenance Contract (CMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

### Recommendation to the User:

Please keep 10-days emergency back-up cylinders. We've calculated your oxygen requirements strictly as sudden extra oxygen requirements can be backed-up by this emergency back-up. We advise our customers to go for 20-30% higher capacity from the average requirement to avoid neck to neck situation in the peak hours.

## Best ROI Guaranteed

***ROI within 6 to 18 months when compared to conventional methods***

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. MEDO is a fully automatic (operator-free operation) that ensures peace of mind.

# We deliver what we commit!!

Thank You

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# Commercial Offer

| S No | Item Code | Item Name | Description | HSN | Qty | Unit Price | GST(%) | Amount |
|---|---|---|---|---|---|---|---|---|
| 1 | MedO 30-1 | MedO 30 | MedO PSA Package | TBD | 1 | 65,90,000.00 | 18.00 | 65,90,000.00 |
| | | | | | | | Sub Total | 65,90,000.00 |
| | | | | | | | GST(%) @18.00 | 11,86,200.00 |
| | | | | | | | Grand Total () | 77,76,200.00 |

**Absstem Industrial Solutions Pvt. Ltd.**
Ground Floor, Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122052
**Contact**: 1800 547 0506 | indsales@absstem.com
**CIN**: U29256HR2022PTC103387

# Terms And Conditions

| | | |
|---|---|---|
| 1. | EXCLUSIONS & CLIENTâS SCOPE | All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clientâs site will be in Client Scope. Additional gas piping, cooling water (if required). |
| 2. | SUPERVISION OF ERECTION & COMMISSIONING | There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you. |
| 3. | P&F | 3% Extra. |
| 4. | Freight | Extra at Actuals. |
| 5. | Insurance | In your Scope. |
| 6. | Payment Terms | 50% Advance with your Prestegious Purchase Order and Balance against PI. |
| 7. | Delivery | 6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability. |
| 8. | Warranty | We hereby confirm that, our equipment shall have Supplierâs warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier. |
| 9. | GST | As per the Commercial Table. |
| 10. | Remarks | . |

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Mohan Reddy

Regional Sales Manager

9289722207

www.absstem.com